

ZHIXIN ZHAO (赵志新)

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EDUCATION

- Tianjin University (985)** Sep. 2022 - Present
PhD Candidate in Software Engineering Tianjin, China
• Advisor: Prof. Wenyu Qu & Prof. Yitao Hu
- Northwest A&F University (985)** Sep. 2018 - Jun. 2022
B.E. in Computer Science and Technology Shaanxi, China
• Overall GPA: 3.81/4.0 (top 5%)

RESEARCH INTERESTS

Serving System for LLMs and DNNs, Machine Learning Systems

PUBLICATIONS

- PAT: Accelerating LLM Decoding via Prefix-Aware Attention with Resource Efficient Multi-Tile Kernel**
Jinjun Yi*, **Zhixin Zhao***, Yitao Hu, Ke Yan, Weiwei Sun, Hao Wang, Laiping Zhao, Yuhao Zhang, Wenxin Li, Keqiu Li.
ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS'26, CCF-A).
- PARD: Enhancing Goodput for Inference Pipeline via Proactive Request Dropping**
Zhixin Zhao, Yitao Hu, Simin Chen, Mingfang Ji, Wei Yang, Yuhao Zhang, Laiping Zhao, Wenxin Li, Xiulong Liu, Wenyu Qu, Hao Wang. *The 21st ACM European Conference on Computer Systems (EuroSys'26, CCF-A)*.
- Harpagon: Minimizing DNN Serving Cost via Efficient Dispatching, Scheduling and Splitting**
Zhixin Zhao, Yitao Hu, Ziqi Gong, Guotao Yang, Wenxin Li, Xiulong Liu, Keqiu Li, Hao Wang. *IEEE International Conference on Computer Communications (INFOCOM'25, CCF-A)*.
- SLOpt: Serving Real-Time Inference Pipeline with Strict Latency Constraint**
Zhixin Zhao, Yitao Hu, Guotao Yang, Ziqi Gong, Chen Shen, Laiping Zhao, Wenxin Li, Xiulong Liu, Wenyu Qu. *IEEE Transactions on Computers (TC'25, CCF-A)*.
- TightLLM: Maximizing Throughput for LLM Inference via Adaptive Offloading Policy**
Yitao Hu, Xiulong Liu, Guotao Yang, Linxuan Li, Kai Zeng, **Zhixin Zhao**, Sheng Chen, Laiping Zhao, Wenxin Li, Keqiu Li. *IEEE Transactions on Computers (TC'25, CCF-A)*.
- ViDA: Lossless VideoQA Acceleration via Selective Sparse Self-Speculation with Parallel Computational Load Management**
Liang Zheng, Yitao Hu, Chen Shen, **Zhixin Zhao**, Bowen Shi, Yuhao Zhang, Laiping Zhao, Wenxin Li, Keqiu Li. *IEEE Transactions on Parallel and Distributed Systems (TPDS'26, CCF-A)*.
- SmartCache: Two-Dimensional KV-Cache Similarity for Efficient Long-Context LLM Decoding**
Chen Shen, Hao Chen, Kaining Hui, **Zhixin Zhao**, Yang Cheng, Yitao Hu, Sheng Chen, Xiulong Liu, Keqiu Li. *IEEE International Conference on High Performance Computing and Communications (HPCC'25)*.
- High-throughput Sampling, Communicating and Training for Reinforcement Learning Systems**
Laiping Zhao, Xinan Dai, **Zhixin Zhao**, Yusong Xin, Yitao Hu, Jun Qian, Jun Yao, Keqiu Li. *IEEE/ACM International Symposium on Quality of Service (IWQoS'23)*.

PROJECTS

- Global GPU Memory Pooling for Multi-LLM Colocation** 2025-present
• Cloud LLM serving typically deploys models on exclusive GPUs, leading to memory underutilization and compute waste under bursty workloads. We design a **multi-LLM colocation system with global GPU memory pooling** for the unified management of weights, KV cache, and activations, integrated with priority-aware scheduling and dynamic offloading to mitigate resource contention and enhance throughput.

Accelerating LLM RL Rollouts via Load-Balanced Scheduling2025-present

- The long-tail effect in LLM RL rollouts causes imbalanced attention workloads and low resource utilization. We develop a **load-balancing scheduling framework** featuring length-affinity distribution across DP instances and KV-splitting to optimize SM utilization. Additionally, a similarity-based Long-Job-First (LJF) strategy is implemented to mitigate tail latency and enhance rollout throughput.

Prediction-based Two-level Request Scheduling for LLM Serving2025-present

- The uncertain LLM output lengths cause load imbalance across instances and frequent request preemption. We develop a **prediction-based scheduling framework** that leverages output length predictor to enable two-level scheduling: cluster-level multi-dimensional load balancing and instance-level SLO-aware scheduling, effectively improving LLM serving goodput.

PAT: Prefix-Aware Attention for LLM Decoding2024-2025

- [\[ASPLOS'26\]](#) LLM serving systems face efficiency bottlenecks during the decoding phase due to heavy KV cache read overhead. We design **PAT (Prefix-Aware Attention)**, which eliminates redundant KV cache reads by reorganizing computation around intra-batch shared prefix structures, achieving substantial speedup in multi-tenant LLM serving scenarios.

Serving Systems for DNNs: Scheduling and Resource Management2023-2025

- DNN serving systems face challenges in maintaining strict latency while minimizing costs in complex inference pipelines. We optimized the end-to-end serving lifecycle through three systems:
- [\[EuroSys'26\]](#) **PARD** optimizes inference pipeline goodput by proactively dropping requests with insufficient latency budgets. It leverages runtime estimation and adaptive prioritization to make timely dropping decisions and mitigate resource waste.
- [\[INFOCOM'25\]](#) **Harpagon** reduces serving costs via batch-aware request dispatching and multi-tuple configuration optimization to improve throughput under latency constraints (i.e., goodput).
- [\[TC'25\]](#) **SLOpt** addresses inter-model dependencies and mitigates cascade blocking through cascade workload estimation, ensuring efficiency and SLO compliance.

SIDE PROJECTS

2025	Daily Arxiv: Daily arXiv paper tracking platform for ML Systems research.	 
2024	Twen: Campus AI assistant in Tianjin University built with RAG.	

COMPETITION AWARDS

2021	The 2021 ICPC Shaanxi National Invitational Contest, Silver Medal
2021	The 2020 ICPC Asia-East Continent Final, Bronze Medal
2021	The 2020 ICPC Asia Shenyang Regional Contest, Bronze Medal
2020	The 45th ICPC Asia Regional Content, Shanghai Site, Silver Medal
2021	LanQiao Cup C/C++ Programming Competition, Group A National Finals, Second Prize
2021	China Collegiate Computing Contest - Group Programming Ladder Tournament, First Prize
2020	China Collegiate Computing Contest - Group Programming Ladder Tournament, First Prize

HONORS AND AWARDS

2023-2025	Academic Scholarships	Tianjin University
2022	Distinguished Academic Scholarship (Top 5%)	Tianjin University
2022	Outstanding Graduates (Top 2%)	Northwest A&F University
2021	Presidential Scholarship	Northwest A&F University
2019-2020	National Encouragement Scholarship	Northwest A&F University